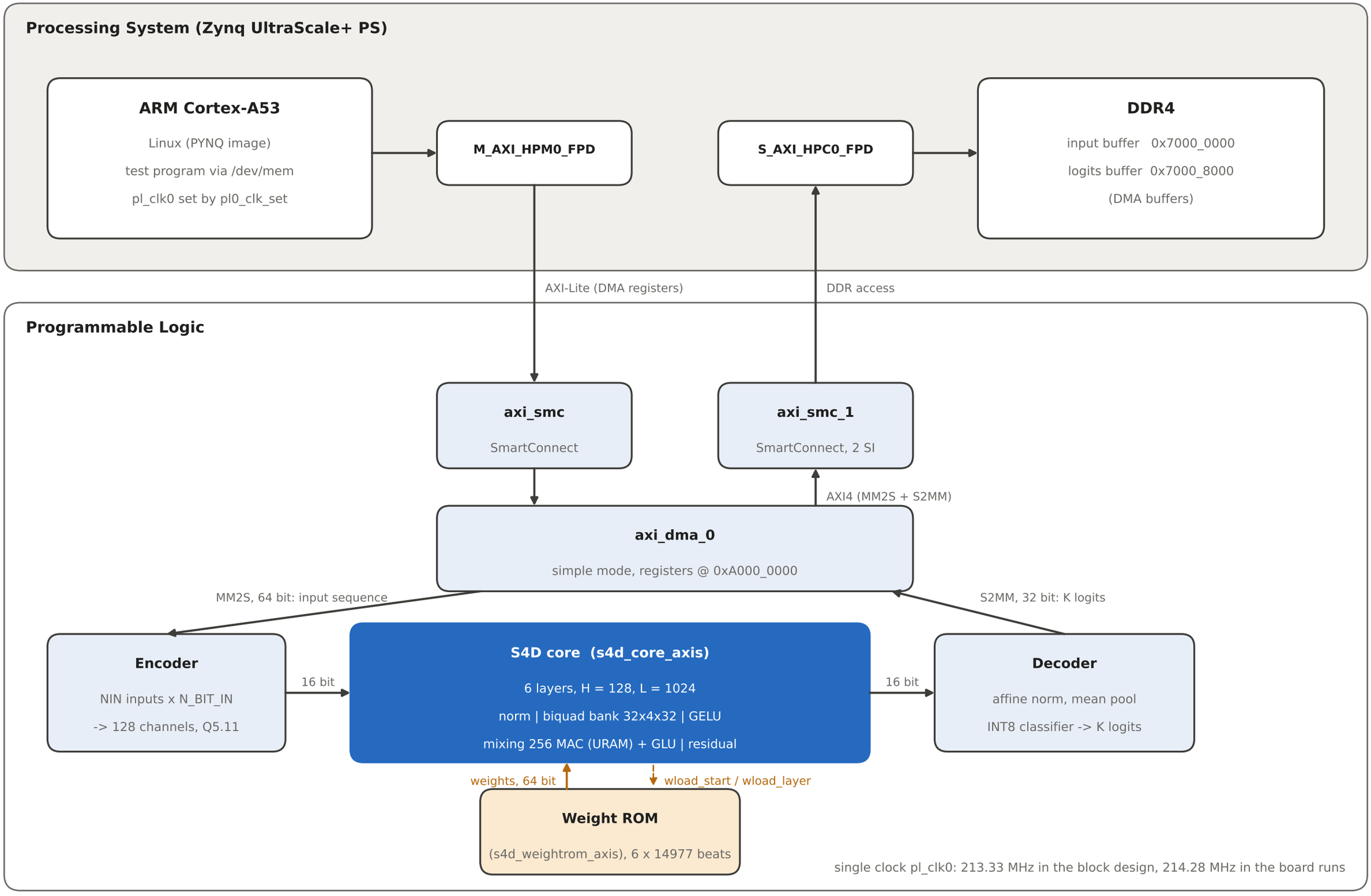
Processing System (Zynq UltraScale+ PS)
ARM Cortex-A53
Linux (PYNQ image)
test program via /dev/mem
pl_clk0 set by pl0_clk_set
M_AXI_HPM0_FPD
S_AXI_HPC0_FPD
DDR4
input buffer 0x7000_0000
logits buffer 0x7000_8000
(DMA buffers)
AXI-Lite (DMA registers)
DDR access
Programmable Logic
axi_smc
SmartConnect
axi_smc_1
SmartConnect, 2 SI
AXI4 (MM2S + S2MM)
axi_dma_0
simple mode, registers @ 0xA000_0000
MM2S, 64 bit: input sequence
S2MM, 32 bit: K logits
Encoder
NIN inputs x N_BIT_IN
-> 128 channels, Q5.11
16 bit
S4D core (s4d_core_axis)
6 layers, H = 128, L = 1024
norm | biquad bank 32x4x32 | GELU
mixing 256 MAC (URAM) + GLU | residual
16 bit
Decoder
affine norm, mean pool
INT8 classifier -> K logits
weights, 64 bit
wload_start / wload_layer
Weight ROM
(s4d_weightrom_axis), 6 x 14977 beats
single clock pl_clk0: 213.33 MHz in the block design, 214.28 MHz in the board runs